Sarvesh Gharat

sarvesh.gharat@iitb.ac.in | sarveshvgharat.github.io | scholar.google/sarvesh-gharat | linkedin.com/sarvesh-gharat

Incoming Visiting Researcher at MBZUAI (2026) — Multi-Agent Systems for Scientific Discovery

Education

IIT Bombay Mumbai, In
PhD in Artificial Intelligence and Data Science *July 2022 – May 2027*

Vishwakarma Institute of Information Technology Pune, In
BTech in Electronics and Telecommunication *Aug. 2018 – May 2022*

Research Interests

My research focuses on online learning, particularly multi-armed bandits and Markov decision processes (MDPs) for sequential decision-making. I'm also interested in multi-agent systems and currently work on inference-time algorithms along with the bandit theory.

Experience

- **Summer Research Intern** May 2025 – July 2025
  Adobe Research *Host: Soumyabrata Pal, Ramasuri Narayanam*
  - Improved LLM reasoning under fixed compute constraints by developing a novel method to inject auxiliary information during sequential test-time scaling.
  - Designed and implemented an efficient, on-the-fly example selection strategy using embedding similarity, leading to more relevant in-context examples during inference.
  - Focused on designing inference-time techniques that enhance model responses without changing any underlying parameters.

Patent Filed: Gharat, S., Pal, S., Narayanam. R., Guided Sequential Test Time Scaling using additional hints from exemplars

- **Student Researcher** Oct 2024 – March 2025
  Google DeepMind *Host: Aparna Taneja, Milind Tambe*
  - Quantified the impact of AI-scheduled interventions on beneficiary engagement and health knowledge in a large-scale maternal health program, analyzing data from thousands of participants.
  - Demonstrated statistically significant improvements in selected maternal health knowledge and postnatal care behaviors via a rigorous comparative analysis of AI-targeted interventions versus matched control groups.
  - Improved reliability of survey-based insights by addressing noise and response variability, enabling statistically significant impact detection.
- **Visiting Researcher** Aug 2021 – June 2022
  Tata Institute of Fundamental Research *Host: Prof. Shriganesh Prabhu*
  - Designed optimization methods to estimate the refractive index of optically thin materials using Terahertz Time Domain Spectroscopy.
  - Implemented and benchmarked algorithms including Dual Annealing, SHGO, and Newton-Raphson to improve convergence stability.
  - Built a user-friendly Python library to support experimental analysis and make the refractive index estimation pipeline accessible for both researchers and educators.

Papers Under Review/ In Preparation

- **Gharat, S.**, Chavan, U., Karamchandani, N. and Nair, J., Which Model to Train? A Budgeted Bandit Approach to Adaptive Fine-Tuning. **Under Review**.
- **Gharat, S.**, Karamchandani, N., and Nair, J., Cost-Aware Best Arm Identification via Dueling Feedback with Applications to Large Language Models. **Full Version under review**.
- Kumar, A., **Gharat, S.**, Beyond One-Shot Decisions: Progressive Morphological Reasoning in Galaxy Classification **Under Review**.
- Singh, V., Ghosal, S., **Gharat, S.**, Pal, S., Narayanam. R., Manocha. D, ThinkRetrieve: Retrieval-Augmented Reasoning Traces for Test-Time Scaling **Under Review**.

Accepted Publications

- **Gharat, S.**, Karamchandani, N. and Nair, J., 2026, May. Cost-Aware Best Arm Identification via Dueling Feedback with Applications to Large Language Models. To appear in Proceedings of the 25th International Conference on Autonomous Agents and Multiagent Systems as an extended abstract.
- Dasgupta, A.*, **Gharat, S.***, Madhiwalla, N., Hegde, A., Tambe, M., Taneja, A. AI-Targeted Calls Drive Measurable Improvements in Maternal Health, PAIS ECAI 2025.
- **Gharat, S.**, Yadav, A., Karamchandani, N. and Nair, J., 2025, April. Representative arm identification: A fixed confidence approach to identify cluster representatives. In ICASSP 2025-2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) (pp. 1-5). IEEE.
- Suresh, R., Bishnoi, H., Kuklin, A.V., Parikh, A., Molokeev, M., Harinarayanan, R., **Gharat, S.** and Hiba, P., 2024. Revolutionizing physics: a comprehensive survey of machine learning applications. Frontiers in Physics, 12, p.1322162.
- Bhatta, G., **Gharat, S.**, Borthakur, A. and Kumar, A., 2024. Gamma-ray blazar classification using machine learning with advanced weight initialization and self-supervised learning techniques. Monthly Notices of the Royal Astronomical Society, 528(1), pp.976-986.
- Mistry, P., Prasad, A., Maity, M., Pathak, K., **Gharat, S.**, Lekkas, G., Bhattarai, S., Kumar, D., Lissauer, J.J., Twicken, J.D. and Soubkiou, A., 2024. VaTEST III: Validation of eight potential super-earths from TESS data. Publications of the Astronomical Society of Australia, 41, p.e030.
- **Gharat, S.**, Borthakur, A. and Bhatta, G., 2024. Estimation of redshift and associated uncertainty of Fermi/LAT extragalactic sources with Deep Learning. Monthly Notices of the Royal Astronomical Society, 527(3), pp.6198-6210.
- **Gharat, S.**, Bose, B., Borthakur, A. and Mazumder, R., 2023. An image processing approach to identify solar plages observed at 393.37 nm by the Kodaikanal solar observatory. RAS Techniques and Instruments, 2(1), pp.393-397.
- Mistry, P., Pathak, K., Prasad, A., Lekkas, G., Bhattarai, S., **Gharat, S.**, Maity, M., Kumar, D., Collins, K.A., Schwarz, R.P. and Mann, C.R., 2023. VaTEST. II. Statistical Validation of 11 TESS-detected Exoplanets Orbiting K-type Stars. The Astronomical Journal, 166(1), p.9.
- Pininti, V.R., Bhatta, G., Paul, S., Kumar, A., Rajgor, A., Barnwal, R. and **Gharat, S.**, 2023. Exploring short-term optical variability of blazars using TESS. Monthly Notices of the Royal Astronomical Society, 518(1), pp.1459-1471.
- **Gharat, S.** and Dandawate, Y., 2022. Galaxy classification: a deep learning approach for classifying Sloan Digital Sky Survey images. Monthly Notices of the Royal Astronomical Society, 511(4), pp.5120-5124.
- Kulkarni, J.S., Cengiz, K. and **Gharat, S.**, 2021, March. Compact C-slot microstrip-fed planar antenna for wireless devices. In 2021 International Conference on Emerging Smart Computing and Informatics (ESCI) (pp. 651-654). IEEE.

Invited Talks & Presentations

- **Invited Talks**
  - Which Model to Train? A Budgeted Bandit Approach to Adaptive Fine-Tuning — Cohere for AI Reinforcement Learning Group (2026)
  - *Cost-Aware Best Arm Identification via Dueling Feedback with Applications to Large Language Models* — Cohere for AI Research Connections (2025)
  - *Representative arm identification: A fixed confidence approach to identify cluster representatives* — Cohere for AI Reinforcement Learning Group (2025)

- **Poster Presentation:**
  - ACM India ARCS 2026
  - Google Research Week 2024
  - IndoML Graduate Forum 2024
  - ACM Pingala Interactions in Computing 2024

Academic Service & Highlights

- **Reviewer / Program Committee**
  - IJCAI AI for Social Good (2025, 2026) - Awarded Bronze Tier reviewer distinction in 2026
  - KDD AI for Sciences 2026
  - AAMAS 2025
  - IEEE Transactions on Information Theory
- **Research Grant**
  - Selected for the *Cohere Labs Catalyst Grant Program* for a proposal on online learning, receiving an initial $1,000 in Cohere API credits with renewable support over one year.
  - Inaugural cohort grantee of the *Adaption Labs Research Grant*, receiving full platform access for research on continual learning, online learning, and adaptive AI systems
  - State Bank of India (SBI) funded project (2024–2026) on *Dynamic LLM Optimization and Fine-Tuning via Dueling Bandits*, exploring adaptive model selection under resource constraints
- **Selected Programs**
  - AAMAS Doctoral Consortium 2026
  - ACM India ARCS 2026
  - Google Research Week (2024, 2025)
  - IndoML Graduate Forum 2024
  - ACM CODS-COMAD PhD Clinic 2024
  - ACM Pingala Interactions in Computing 2024
- **Kaggle Expert**
  - LLM Prompt Recovery Challenge: Silver Medal (Rank 79/2175)
  - Santa 2025 Challenge: Bronze Medal (Rank 228/3357)

Teaching Experience

- **CS 747:** Foundations of Intelligent and Learning Agents **(Best TA Award)**
- **CS 663:** Digital Image Processing
- **EE 325:** Probability and Random Processes
- **DS 203:** Programming for Data Science

Technical Skills

- **Programming**: Python, LaTeX, Apps Script
- **Libraries**: PyTorch, TensorFlow, NumPy, pandas, vLLM, Transformers
- **Tools**: Jupyter Notebook, Git, Docker, Google Cloud Platform, VS Code